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Graduation Project Proposal

Project Title	Grow Lead: AI-Powered Personal & Leadership Development Platform for Students
Project Summary	University personal and leadership development programs (LDPs) currently utilize a manual, fragmented set of processes such as journals, photos, forms, and spreadsheets. Not only does this diminish the student's engagement, depth of reflection, and ability to track personal progress, it creates a burden on professors and offers little in the way of analytics to the universities offering these courses. Grow lead is an online database that houses a suite of tools that seek to unify a student's developmental journey, from goal setting, logging of activities (journaling, reading, volunteer, leadership), submission of evidence, and artificial intelligence powered recommendations. Centralized dashboards for professors and institution-wide analytics for universities. The final product will be a continuous self-development process that will allow students to both track and understand their growth, limiting the burden on instructors while granting universities valuable student behavioral analytics.
Expected Deliverables	SW (O), HW ( ), Patent ( ), Paper (O), Program Registration ( )
Advisor	Ma'am Naira and Ma'am Faiqa
Expected Period	2026.09.15 – 2027.02.15

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Project Background and Objectives

Grow Lead is an AI-powered personal and leadership development platform engineered specifically for university students to digitize, unify, and elevate non-academic self-improvement programs. By integrating daily behavioral tracking with formal mentor evaluations, the platform transforms fragmented personal logs into actionable developmental insights. Grow Lead covers the complete student development lifecycle through six interconnected phases: setting targeted goals across multi-dimensional growth areas, logging daily reflections, reading hours, and service activities via a mobile interface, uploading digital evidence artifacts like certificates and photo logs, routing submissions to an instructor web dashboard for formal scoring and qualitative feedback, calculating a multi-dimensional Personal Development Index (PDI) through a Python analytics engine to quantify progress, and utilizing AI-driven pattern detection to generate personalized next actions, habit reminders, and workload management guidance.

Technical background: University Leadership Development Programs (LDPs) currently rely on a fragmented patchwork of manual and semi-digital processes to track student activities, forcing students to split their records across personal journals, photos, messaging apps, google forms, and spreadsheets. While consumer habit-tracking applications promote personal consistency, they lack integration with university-level evaluation frameworks, reflective journaling, and long-term developmental intelligence loops. Similarly, standard task-oriented productivity tools manage upcoming academic deadlines but fail to capture physical evidence of leadership activities or provide intelligence specifically tuned to personal and soft-skill growth. Furthermore, specialized leadership software often focuses strictly on administrative tracking without offering a comprehensive blend of behavioral analytics, instructional evaluation mechanisms, and academic workload management, leaving a significant gap for an all-in-one platform like Grow Lead.

Necessity: The necessity of the Grow Lead platform stems from fundamental inefficiencies affecting all key stakeholders in traditional university Leadership Development Programs (LDPs): students

complete required tasks mechanically without gaining meaningful insight into their long-term growth trends, strengths, or weak areas; professors waste significant time manually collating physical records, verifying evidence, and scoring entries rather than mentoring; and universities lack a centralized, bird's-eye view of student participation rates and overall developmental outcomes. To resolve these core gaps, this project investigates how AI and behavioral analytics can provide personalized recommendations and actionable intelligence to systematically improve university students' personal and leadership development.

Final goal: The final goal of Grow Lead is to transform personal development management from a laborious, manual record-keeping process into a scientific, continuously evolving, and easily managed system. By centralizing activity tracking, digital journaling, and evidence verification into a single platform, students gain immediate, real-time visibility into their growth trends and receive actionable, AI-driven recommendations on what steps to take next. Concurrently, professors are relieved of heavy administrative tasks—such as manually collecting and tracking physical records—allowing them to focus on high-impact mentoring and feedback using consolidated digital evidence. Ultimately, this unified approach empowers universities to seamlessly digitize, scale, and evaluate their Leadership Development Programs (LDPs) using central, data-driven analytics.

Detailed goals: (1) Cross-platform student mobile app (Flutter + Dart) for goals, journal, reading, service/leadership logs and evidence upload; (2) Instructor web dashboard for submission review, scoring, feedback and monitoring; (3) Personal Development Index (weighted score across goal achievement, mental/physical development, reading, leadership, community service and reflection); (4) AI analytics engine (Python, NLP/LLM) for pattern detection and personalized recommendations; (5) Productivity features linking LDP tasks with academic deadlines, reminders and priority management; (6) Validation via system testing, usability studies and evaluation of recommendation quality.

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Main Content of the Project

Requirements: Functional requirements include user management (registration, authentication, and roles), development progress (goals, activities, journal, reading, reflection, service), evidence and evaluation (evidence upload, evaluation, approval, scoring, feedback, reports), AI and analytics (dashboard, Personal Development Index, journal, patterns, recommendations), and productivity (scheduling, reminders, deadlines, prioritization, and workload guidance). Non-functional requirements involve security (secure authentication and evidence storage), performance (rapid dashboards), usability (mobile access), reliability, scalability, and privacy-focused, explainable AI recommendations.

Research on fundamental technologies: The architecture was built using Flutter + Dart to develop a student mobile application across platforms, Firebase (Authentication, Firestore, Storage and Cloud Functions) as the backend, Python (Pandas, NumPy and NLP/LLM APIs) for AI/analytics and recommendation logic, and a web dashboard for instructors. The coding was done in VS Code and Android Studio with Git and GitHub for version control. There is a body of literature and tools that address some aspects of student engagement and performance, but there is a research gap in terms of holistic student development, instructor evaluation, behavioral analytics and personalized next actions.

R&D trends: Current trends in educational technologies focus on personalized learning, learning analytics, artificial intelligence coaching, and mobile technologies. Studies have shown that through dashboards it is possible to increase self-awareness, using digital tracking to increase rates of engagement, and through centralized evaluations reduce the workload. Yet, few systems aim to combine those to offer a comprehensive solution for executive development for university leadership. Grow Lead offers to take a step further and include a Personal Development Index as well as use artificial intelligence to detect patterns and recommend optimal workloads.

Specific R&D plans: Core modules will be developed iteratively — authentication and profiles, goal and activity tracking, digital journal and reading logs, evidence upload with instructor review, analytics dashboards and Personal Development Index, AI recommendation engine, and productivity/reminder features. Data models will support role-based access. AI components will use behavioral pattern detection and LLM APIs for explainable suggestions. Prototype will be validated through unit, system, usability and AI quality testing with target users.

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Implementation Plan

Month 1 (Weeks 1–4) – Research & Requirements

- Weeks 1–2: Finalize problem definition, literature review on learning analytics, self-regulated learning, leadership development programs and existing digital tools. Prepare research questions (RQ1–RQ3).
- Weeks 3–4: Stakeholder interviews and questionnaire with students and LDP instructors, analyse responses, produce Software Requirements Specification (SRS).

Month 2 (Weeks 5–8) – Design & Early Development

- Weeks 5–6: System design – UI/UX wireframes & high-fidelity prototypes (student mobile app + instructor web dashboard), Fire store database schema, architecture diagrams, UML use-case and sequence diagrams.
- Weeks 7–8: Start core development – authentication & user profiles (Firebase Auth + role-based access), basic goal-setting and activity logging modules.

Month 3 (Weeks 9–12) – Core Feature Development

- Weeks 9–10: Digital journal, reading tracker, community service & leadership activity logs, evidence upload with media storage.
- Weeks 11–12: Instructor web dashboard (review, scoring, feedback, monitoring) + student progress dashboard with basic analytics.

Month 4 (Weeks 13–16) – AI, Analytics & Testing

- Weeks 13–15: Personal Development Index (weighted scoring), pattern detection, AI recommendation engine (Python + NLP/LLM APIs), reminders and workload prioritization.
- Week 16: Internal system testing, bug fixing, unit & integration testing.

Month 5 (Weeks 17–20) – Evaluation & Delivery

- Week 17: Usability testing with students and instructors + AI recommendation quality evaluation.
- Weeks 18–19: Incorporate feedback, final refinements, complete documentation (user manual, technical report, test reports).
- Week 20: Final evaluation, project presentation, demonstration and submission of all deliverables.

Team composition: This is an individual final-year project led by Kinzul Eman (BS Computer Science). The only student kinzul eman is responsible for all aspects: requirements analysis, system design, full-stack development (Flutter frontend, Firebase backend, Python AI components, instructor web dashboard), testing, documentation and presentation. External feedback may be sought from LDP instructors and potential student users during evaluation phases.

Conclusion

Expectations for deliverables: A prototype of the Grow Lead platform (student mobile app + instructor web dashboard) featuring basic tracking, evidence, evaluation, analytics, and AI recommendation functions. Associated documentation, including SRS, design artifacts, testing, and user evaluation reports. A paper describing the research findings regarding the application of AI in personal development.

Individual capabilities expected to improve: Full-stack mobile and web development skills (Flutter, Firebase, Python), system design and architecture, requirements engineering, AI/ML for recommendations and analytics, usability testing and evaluation methodology, technical writing and research communication, project planning and time management under academic constraints.

Application fields and expected effects: Indirectly applicable to university personal development, leadership development programs (LDPs), student affairs and mentorship systems. Academic contribution: empirical insight into whether AI dashboards and recommendations improve student self-awareness, activity consistency and instructor efficiency (addressing RQ1-RQ3). Industrial/educational impact: a model for digitizing holistic student development, which can be adopted and scaled across universities as a productivity layer to cut costs and inform student success strategies. Future work could involve calendar/wearable sync, voice journaling, language localization and institutional analytics.

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References

The following sources informed the problem analysis, technology choices and research framing of this proposal.

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